



Summative User Study

Study design, instruments, evidence gate, and analysis plan

STUDY STATUS

BLOCKED_PARTICIPANT_COUNT · No verified participant sessions. No participant findings are inferred.

The report reflects the canonical analysis result: BLOCKED_PARTICIPANT_COUNT. It changes when evidence is ingested and the analyzer is rerun; no participant findings are inferred from empty or synthetic templates.

Study objective and research questions

Measure whether the selected hi-fi flows support effective, efficient, low-error, and satisfactory completion of the core tasks. The study evaluates interface behavior, not the participant.

- Can users identify FIFA status, ownership, and the next action without coaching?
- Can users recognize the partner boundary and return with context?
- Can users identify and explain the Chess mistake before advanced analysis?
- Can users perform a safer Chess move, recover, and complete related practice?
- What errors, pauses, wording problems, and recovery behaviors remain?

Canonical analysis state

Evidence item		Current state	Required before PASS
Verified participants		No verified participant sessions	At least five verified, anonymized participant sessions
Recordings		No verified video recordings	One consented MP4 with positive duration and a video stream per session
Task results		No task rows (header-only template)	All assigned task rows with timestamps, 2/1/0 score, recovery, and hesitation fields
Questionnaire		No questionnaire rows (header-only template)	Five raw items per product and verified participant
Interview		No interview/feedback rows (header-only template)	At least one timestamped non-empty feedback row per verified participant

Gate	Name	Status
G01	participant count	BLOCKED_PARTICIPANT_COUNT
G02	participant metadata completeness	BLOCKED_PARTICIPANT_METADATA
G03	recording presence and validity	BLOCKED_RECORDINGS
G04	task-result completeness	BLOCKED_TASK_DATA
G05	questionnaire completeness	BLOCKED_QUESTIONNAIRE
G06	interview/feedback completeness	BLOCKED_INTERVIEW_DATA
G07	quantitative-analysis readiness	BLOCKED_EXTERNAL_EVIDENCE
G08	final summative-report readiness	BLOCKED_EXTERNAL_EVIDENCE

Setup and order-neutral procedure

- Quiet room or call; desktop 1440 × 900 and mobile 390 × 844 where appropriate.
- Moderated concurrent think-aloud with descriptive task time and neutral prompts only.
- Observer records first action, timestamps, errors, wrong paths, assistance, recovery, and hesitation.
- Recording begins after consent and is verified using the media rule below.
- Stop for withdrawal, discomfort, recording failure, or coaching risk.

Step	Activity
1	Greeting, purpose, consent, and recording confirmation.
2	Background questions and think-aloud practice.
3	Run the first assigned product tasks, then that product's questionnaire.
4	Reset the prototype.
5	Run the second assigned product tasks, then that product's questionnaire.
6	Neutral post-test interview, closing, and recording filename confirmation.

Counterbalancing

- A_FIFA_FIRST: FIFA tasks → FIFA questionnaire → reset → Chess tasks → Chess questionnaire.
- B_CHESS_FIRST: Chess tasks → Chess questionnaire → reset → FIFA tasks → FIFA questionnaire.

Tasks and measures

Task	Success endpoint	Measures
FIFA-T1	Status, owner, timing, and next action explained	2/1/0; time; errors; wrong paths; assistance; hesitation
FIFA-T2	Order or ticket detail opened and interpreted	Same task measures
FIFA-T3	Partner destination recognized before leaving	Same measures; boundary recognition
FIFA-T4	Return restores status context	Same measures; orientation
CHESS-T1	Beginner Review started independently	Same task measures
CHESS-T2	Mistake and immediate consequence explained	Same measures; own-words explanation
CHESS-T3	Safer move performed and feedback understood	Same measures; source-to-destination interaction; retry/recovery
CHESS-T4	New practice position completed and context retained	Same measures; practice input and continuation

2 = independent completion without facilitator assistance; 1 = completion after one neutral prompt or a defined recoverable wrong path; 0 = failure, abandonment, or solution-revealing/direct assistance.

Track hesitation separately: one event is at least 5 consecutive seconds without task-progress action while the participant is visibly attending, excluding expected reading. Store hesitation_count, timestamp, and observable behavior; hesitation alone does not lower success_score.

Questionnaire and interview

Use the custom five-item Likert questionnaire after each assigned product flow. Store raw 1–5 responses and do not replace missing answers. Ask the neutral interview questions and create only timestamped, non-empty evidence-backed feedback rows.

ID	Statement
Q1	I could find the first useful action without help.
Q2	The status or explanation was clear.
Q3	I felt confident about what to do next.

ID	Statement
Q4	The feedback after my action was useful.
Q5	I was satisfied with this flow.

Recording and analysis rule

A verified recording is an existing MP4 with a positive duration and at least one codec_type=video stream. A present file without a probe is RECORDING_PRESENT_UNVERIFIED and never counts as verified.

Limitations

- The prototype uses static fictional data and does not measure live service reliability or partner completion.
- Participant-dependent findings remain subject to the canonical analysis gates shown above.
- PA3 formative evidence is continuity evidence and is not substituted for PA4 human sessions.
- Concurrent think-aloud timings are descriptive and are not a natural unmoderated baseline.

Evidence index

Artifact	Purpose	State
study-plan.md	Design, criteria, procedure, tasks, measures, privacy	Prepared
facilitator-script.md	Neutral moderated session script	Prepared
data/*.csv	Raw evidence capture schema	Header-only until real evidence is ingested
evidence/recordings/README.md	Recording convention and video verification rule	Prepared; no recordings claimed
study/analysis/	Canonical gates, result, metrics, and synthetic-only tests	Generated from current evidence state

NEXT REQUIRED ACTION

Run the prepared study, verify consented video recordings and timestamped feedback, ingest real evidence, rerun the analyzer, and rebuild the reports before any submission package is generated.